

Final Comparison of Best models of the Algorithms

Algorithm	Best Model Version	Accuracy	Precision	Recall	F1-Score	Remarks
K-Nearest Neighbors (KNN)	KNN_6 (Manual K Exploration with rfe_pca_df)	0.8424	0.8512	0.8860	0.8682	Stable and consistent; improved with tuning
Logistic Regression (LR)	LR_6 (Manual K Exploration with rfe_pca_df)	0.8452	0.8595	0.8793	0.8693	Matches KNN performance closely
Artificial Neural Network (ANN)	ANN_2	0.8483	0.8573	0.8888	0.8727	Best recall and F1-score overall
Decision Tree (DT)	DT_2 (GridSearchCV with chi2_pca_df)	0.8317	0.8407	0.8780	0.8589	Moderate performance
Support Vector Machine (SVM)	SVM_1/SVM_2 (rfe_pca_df)	0.8491	0.8488	0.8491	0.8483	Good accuracy; balanced results
Random Forest (RF)	RandomForest_Tuned_RFE	0.8481	0.86	0.89	0.87	Strong performance and robustness

Final Model Summary

After evaluating multiple algorithms — including K-Nearest Neighbors (KNN), Logistic Regression, Decision Tree, Support Vector Machine (SVM), Artificial Neural Network (ANN), and Random Forest — the Tuned Random Forest model (using RFE PCA data frame) was selected as the final model for predicting student depression.

The model achieved an accuracy of 84.81%, precision of 86%, recall of 89%, and an F1-score of 87%, showing a strong balance between accuracy and generalization.

Although the ANN model recorded the highest F1-score (0.8727), Random Forest was chosen due to its higher interpretability, robustness to overfitting, faster training time, and alignment with ethical AI practices through clear feature importance insights.

The tuning process notably improved Random Forest's recall (+10%) and F1-score (+7.4%), effectively reducing false negatives — crucial in identifying students at risk of depression. Its ensemble structure handles both numerical and categorical features, adapts to non-linear relationships, and performs consistently under cross-validation.

Therefore, considering performance, interpretability, and ethical alignment, the Tuned Random Forest (RFE PCA) model is recommended as the most suitable and deployable model for the Student Depression Prediction System.